Ghazi University D.G. Khan

Assignment

Submitted from:-

Name:- Shifa Zahra

Section:- 'E'

Subject:- Programming Fundamental

Semester:- 1st semester

Department:- CS

Morning.

Submitted To:-

Sir Asad Majeed

Scanned with CamScanner

Lab no 3

Task 1: single cin statement

```
#include<iostream>
using namespace std;
int main(){
    int age;
    double pay;
    char section;

    cout<<"enter age, pay and section:";
    cin >> age>>pay>>section;
    cout<<age<<"  "<<pay<<"  "<<section;
    return 0;

}
```

output

Enter age, Pay and section

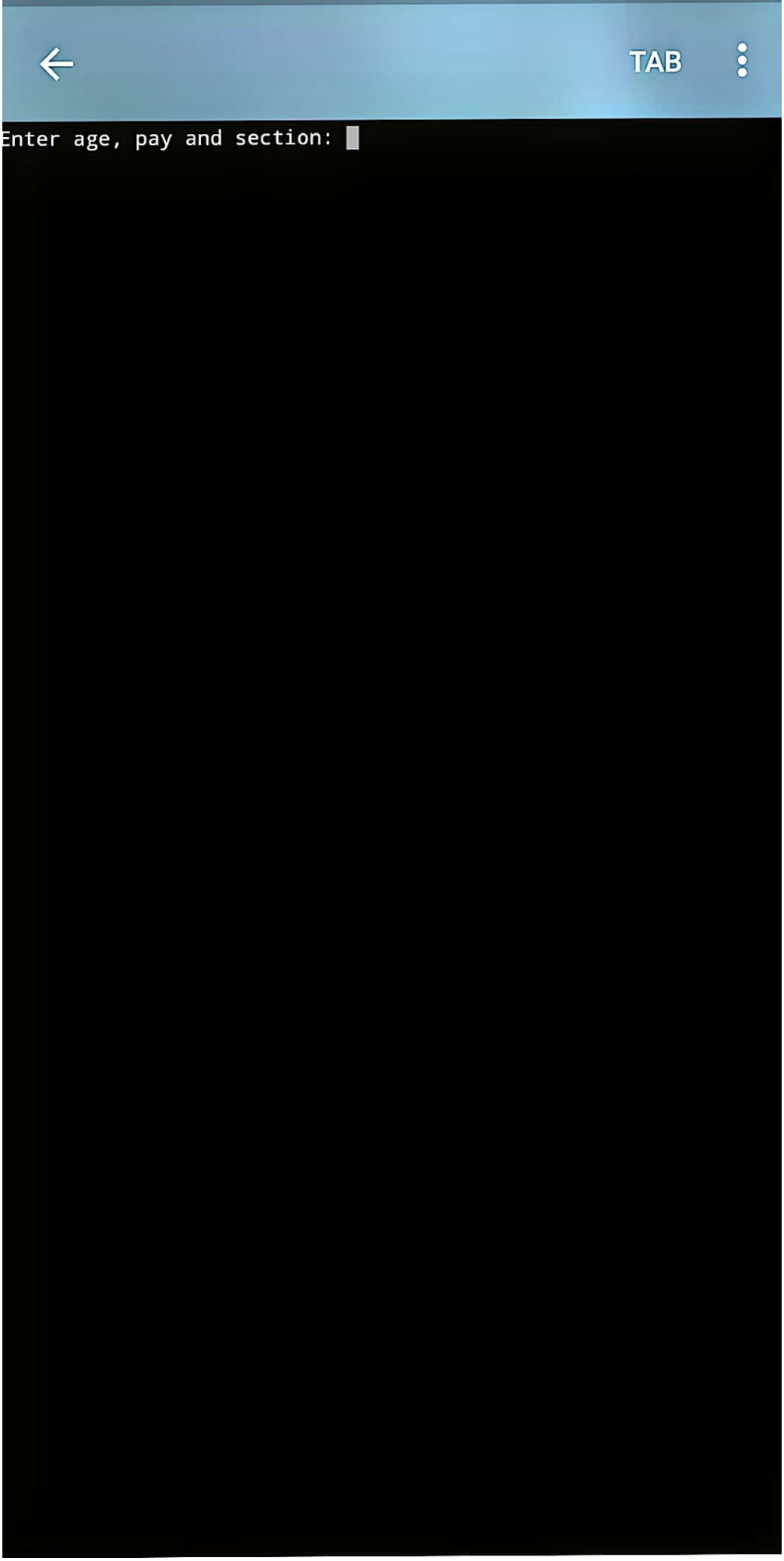
Scanned with CamScanner

new*

```
1  #include <iostream>
2  using namespace std;
3  int main() {
4     int age;
5     double pay;
6     char section;
7
8     cout << "Enter age, pay and section: ";
9     cin >> age >> pay >> section; // single cin
   statement
10    cout << age << " " << pay << " " << section;
11    return 0;
12 }
```

Tab | { | } | : | ; | "

Scanned with CamScanner



Scanned with CamScanner

Task2: values

Expression		Value
28/4-2	=	5
6+12*2-8	=	22
4+8*2	=	20
6+17%3-2	=	6
2+22*(9-7)	=	46
(8+7)*2	=	30
(16+7)%2-1	=	0
12/(10-6)	=	3
(19-3)*(2+2)/4	=	16

Scanned with CamScanner

Task: 3

Correct option is: **C option**

units is promoted to a float, mass remains a float, and the result of mass * units is promoted to a double to be stored in weight

Scanned with CamScanner

Task 4

Given:

```
int a, b = 2;
float c = 4 - 2;
a = b * c;
```

Answer: B(8)

explanation:

1- b-c means 2-4-2 => 8-4-The result is 8.4 of type float.

2- This float value 8.4 is assigned to a, which is an int.

3- when a float is assigned to an int, the decimal part is truncated (removed), not rounded

4- so 8.4 becomes 8.

Therefore, the value stored in a will be 8.

Scanned with CamScanner

Task:5

Given: unitsEach = qty / salesReps;
where qty and salesReps are both int

Problem: This will perform integer division, so decimal part will be lost.

Solution using type cast:

To avoid integer division, we need to cast one operand to float or double.

Correct Answer:

unitsEach = static_cast<double>(qty)/salesReps;

Scanned with CamScanner

Task 6: Math Tutor Program (rand)

```
#include<iostream>
#include<cstdlib>
#include<ctime>
using namespace std;
int main( ){
    srand(time(0));
    int num1=rand( )%900+100;
    int num2=rand(0)%900+100;
    Cout<<"  "<<num1<<endl;
    cout<<"+ "<<num2<<endl;
    Cout<<"-----"<<endl;
    cout<<"\n Press enter to see answer...!!";
    cin.get( );
    cin.get( );
    Cout<<"\n "<<num1<<endl;
    cout<<"+ "<<num2<<endl;
    Cout<<"-----"<<endl;
    Cout<<"  "<<num1+num2<<endl;
    return 0;
}
```

output:-

```
 846
+204
Press enter to see answer----
```

Scanned with CamScanner

new*

```cpp
#include <iostream>
#include <cstdlib> // for rand()
#include <ctime>   // for time()
using namespace std;

int main() {
    srand(time(0));
    int num1 = rand() % 900 + 100;
    int num2 = rand() % 900 + 100;

    cout << "  " << num1 << endl;
    cout << "+ " << num2 << endl;
    cout << "-------" << endl;

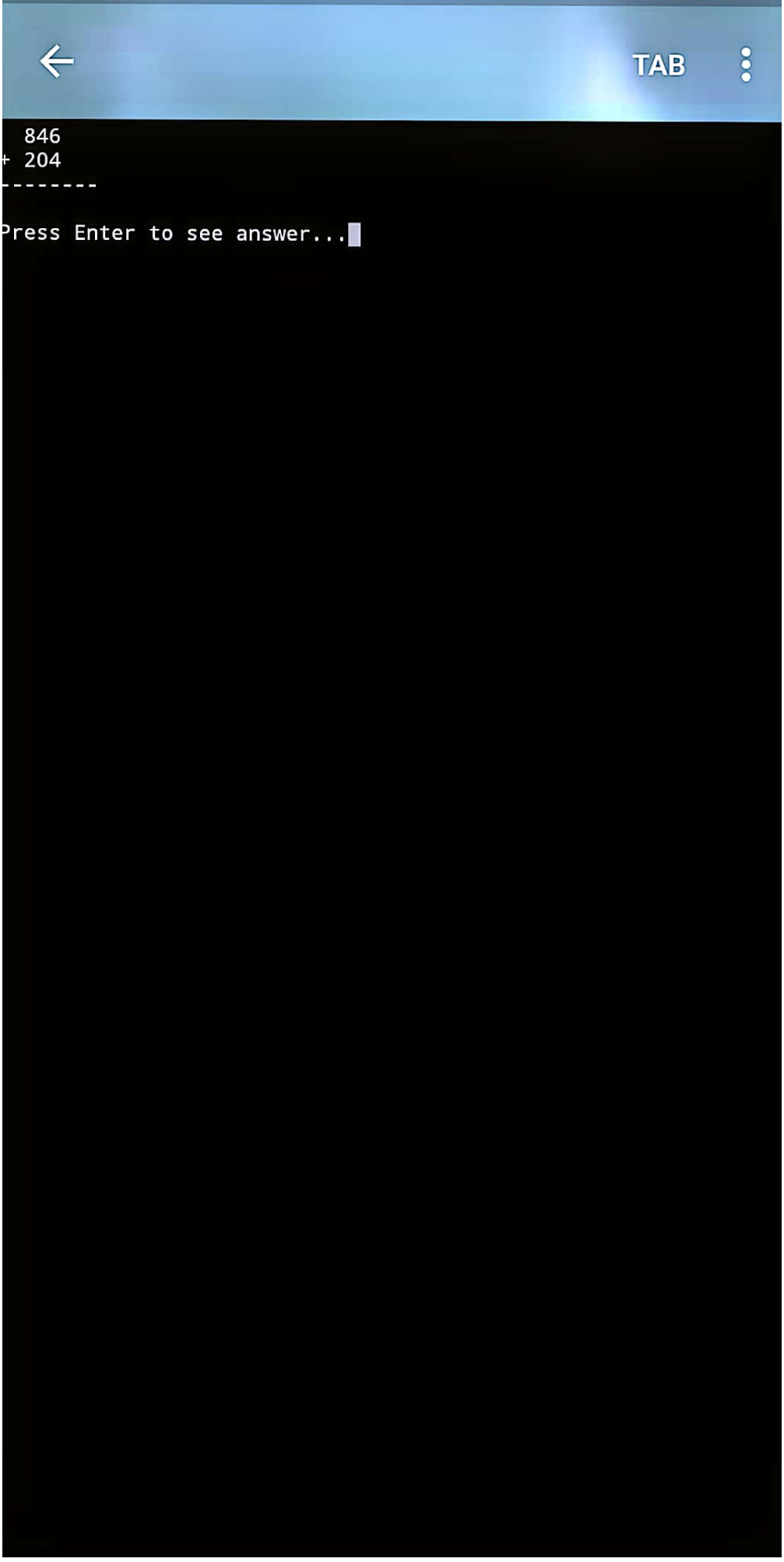
    cout << "\nPress Enter to see answer...";
    cin.get();
    cin.get();

    cout << "\n  " << num1 << endl;
    cout << "+ " << num2 << endl;
    cout << "-------" << endl;
    cout << "  " << num1 + num2 << endl;

    return 0;
}
```

Tab | { | } | : | ; | "

Scanned with CamScanner



Scanned with CamScanner

Lab # 4

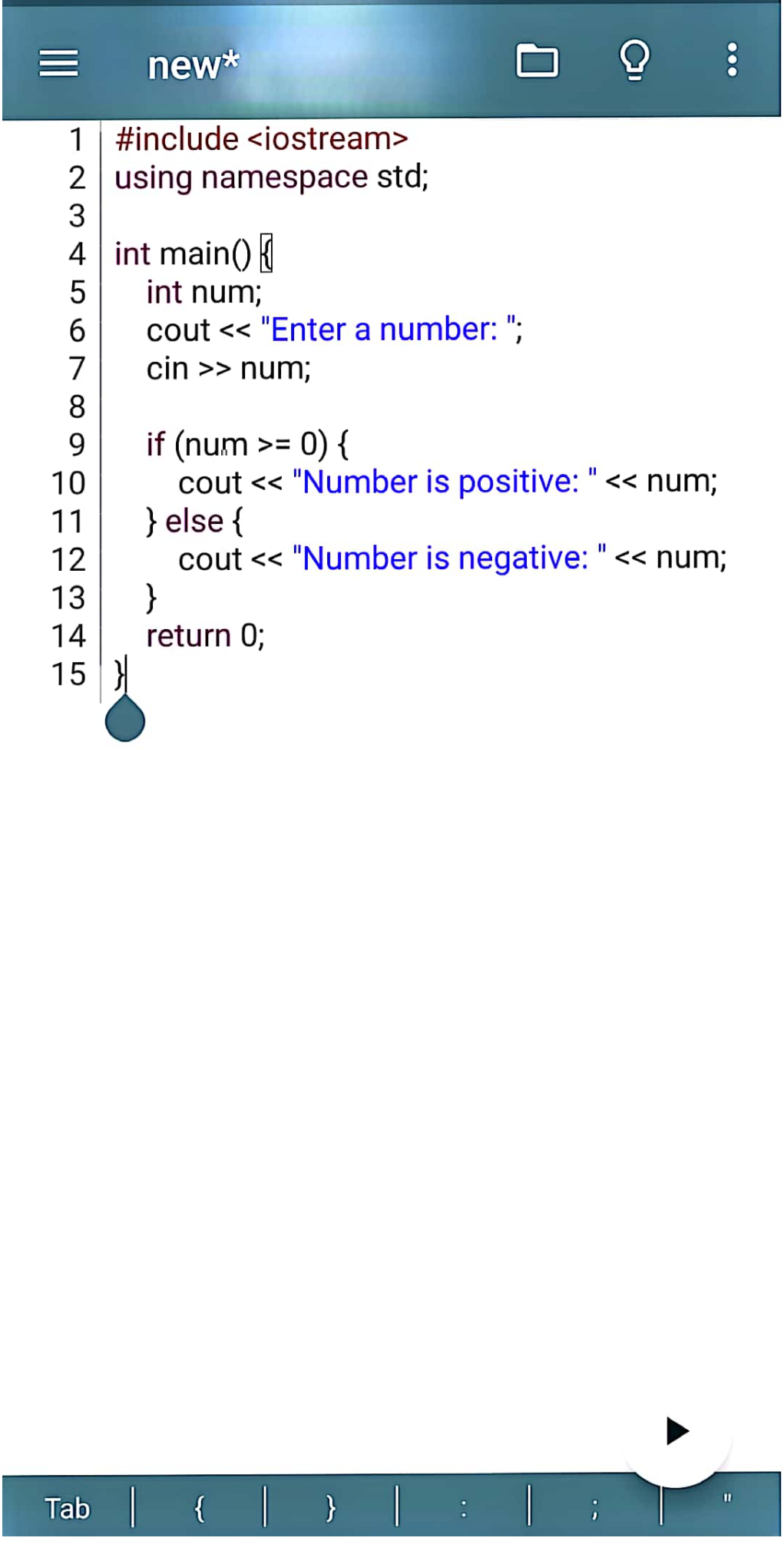
TASK 1:
Positive/Negative

```
#include<iostream>
using namespace std;
int main(){
    int num;
    cout<<"Enter a Number:";
    cin>>num;
    if(num>=0){
        cout<<"Number is Positive"<<num;
    }else{
        cout<<"Number is negative:"<<num;
    }
    return 0;
}
```

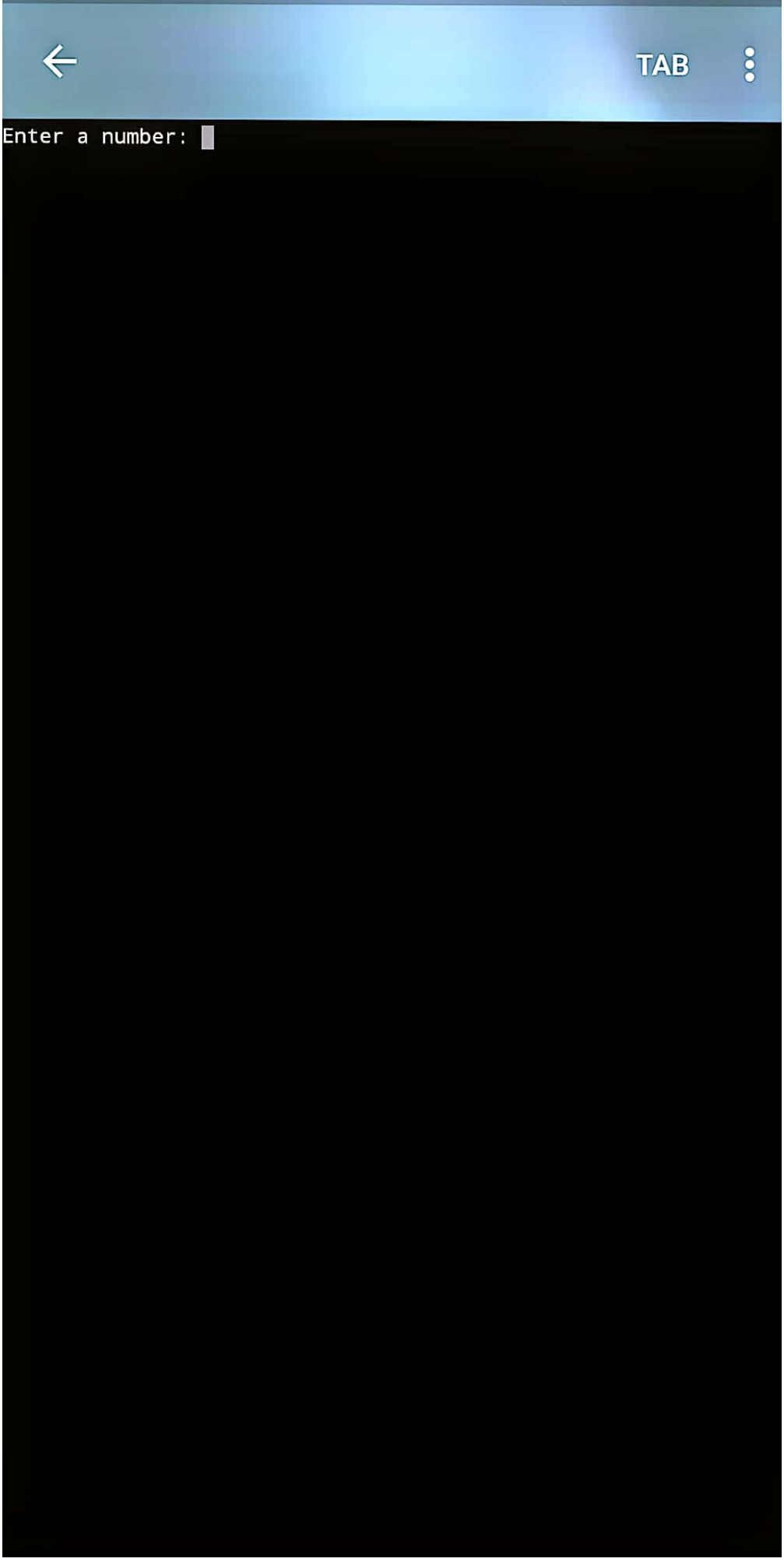
Output

Enter a number

Scanned with CamScanner



Scanned with CamScanner



Scanned with CamScanner

Task 2

Positive, Negative or zero

```
#include<iostream>
using namespace std;
int main(){
    int num;
    if(num>0){
        cout<<"number is Positive";
    }else if(num<0){
        cout<<"number is negative";
    }else{
        cout<<"number is Zero";
    }
    return 0;
}
```

output

Enter an integer

Scanned with CamScanner

new*

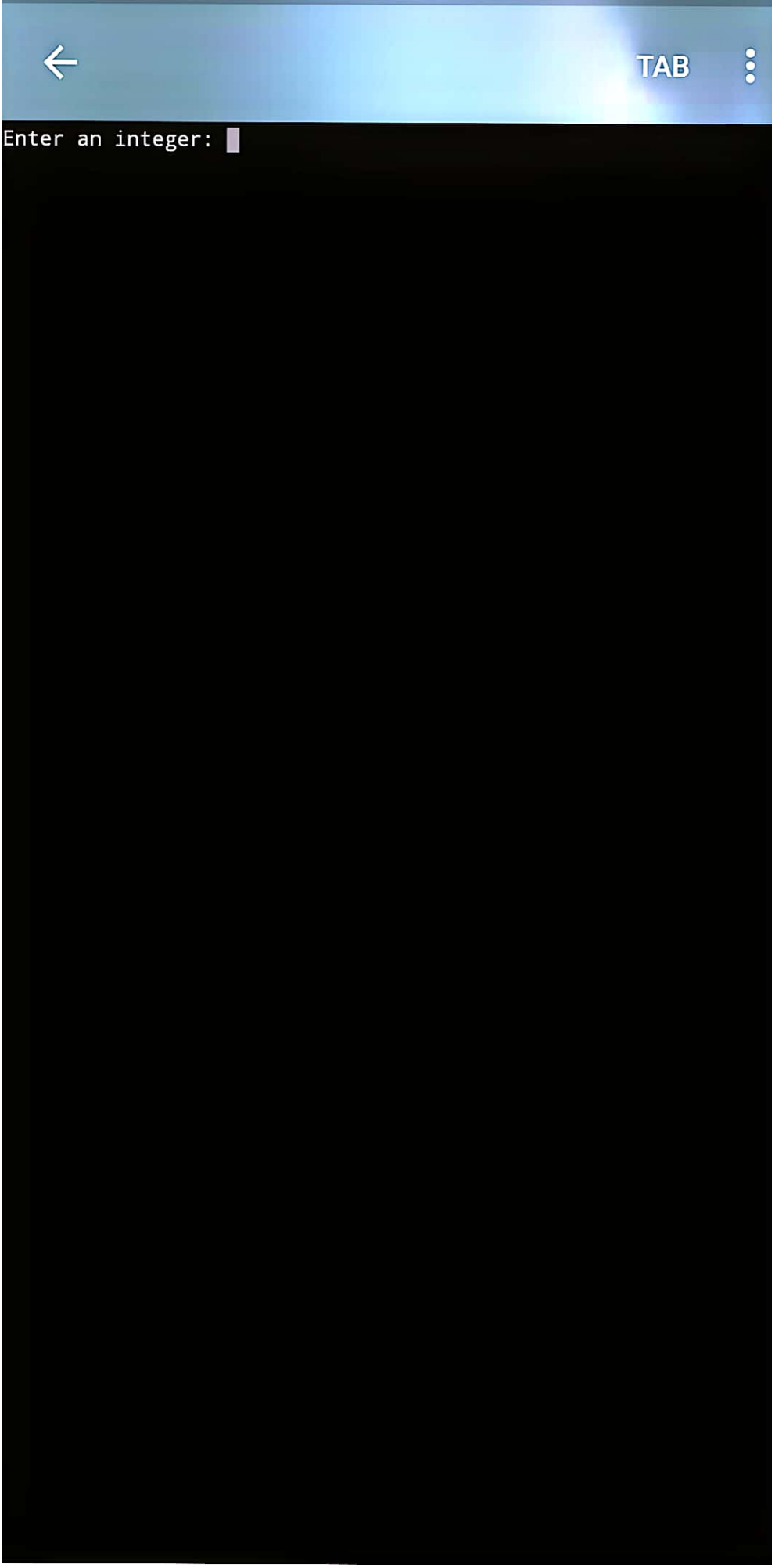
```cpp
#include <iostream>
using namespace std;

int main() {
    int num;
    cout << "Enter an integer: ";
    cin >> num;

    if (num > 0) {
        cout << "Number is positive";
    } else if (num < 0) {
        cout << "Number is negative";
    } else {
        cout << "Number is zero";
    }
    return 0;
}
```

Tab | { | } | : | ; | "

Scanned with CamScanner



Scanned with CamScanner

Task 3

Grand system

```
#include<iostream>
using namespace std;
int main(){
    int mark;
    cout<<"Enter your marks=";
    cin>>mark;
    if(mark>=75){
      cout<<"Grade A";
    }else if(mark>=60){
      cout<<"Grade B";
    }else if(mark>=45){
      cout<<"Grade C";
    }else{
      cout<<"Grade D";
    }return 0;
}
```

output:

Enter your marks

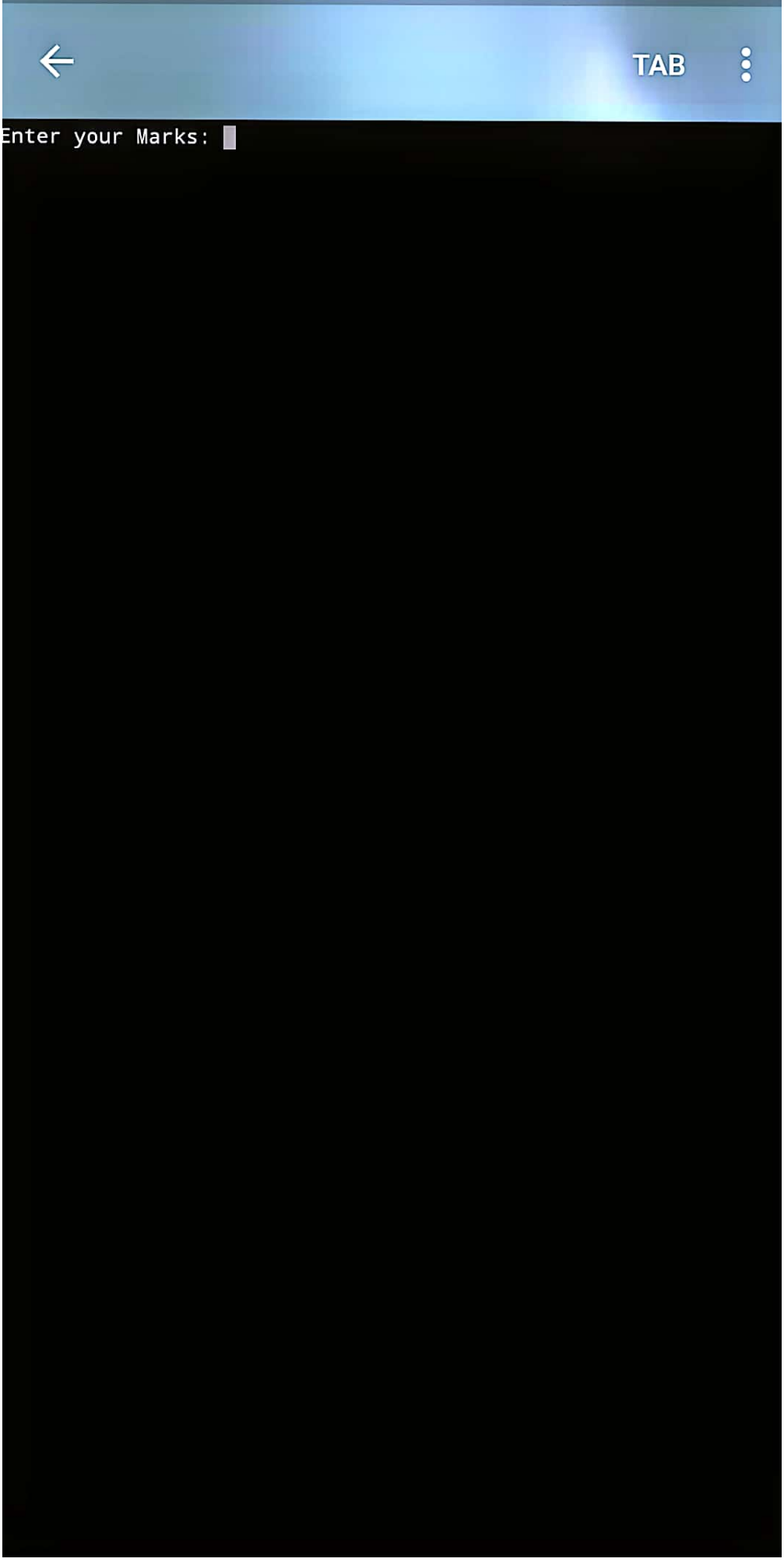
Scanned with CamScanner

new*

```
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int mark;
6      cout << "Enter your Marks: ";
7      cin >> mark;
8
9      if (mark >= 75) {
10         cout << "Grade A";
11     } else if (mark >= 60) {
12         cout << "Grade B";
13     } else if (mark >= 45) {
14         cout << "Grade C";
15     } else {
16         cout << "Grade D";
17     }
18     return 0;
19 }
```

Tab | { | } | : | ; | "

Scanned with CamScanner



Scanned with CamScanner

Task 4

Largest Number + Digits count

A) largest number using Nested if---else

```
#include<iostream>
using namespace std;
int main(){
    int a,b,c;
    cout<<"Enter three numbers:";
    cin>>a>>b>>c;
    if(a>=b){
      if(a>=c){
        cout<<"Largest is:"<<a;
      }else{
        cout<<"Largest is:"<<c;
      }
    }else{
        if(b>=c){
          cout<<"Largest is:"<<b;
        }else{
          cout<<"Largest is:"<<c;
        }
    return 0;
}
```

Scanned with CamScanner

new*

```cpp
#include <iostream>
using namespace std;

int main() {
    int a, b, c;
    cout << "Enter three numbers: ";
    cin >> a >> b >> c;

    if (a >= b) {
        if (a >= c) {
            cout << "Largest is: " << a;
        } else {
            cout << "Largest is: " << c;
        }
    } else {
        if (b >= c) {
            cout << "Largest is: " << b;
        } else {
            cout << "Largest is: " << c;
        }
    }
    return 0;
}
```

Tab | { | } | : | ; | "

Scanned with CamScanner



Scanned with CamScanner

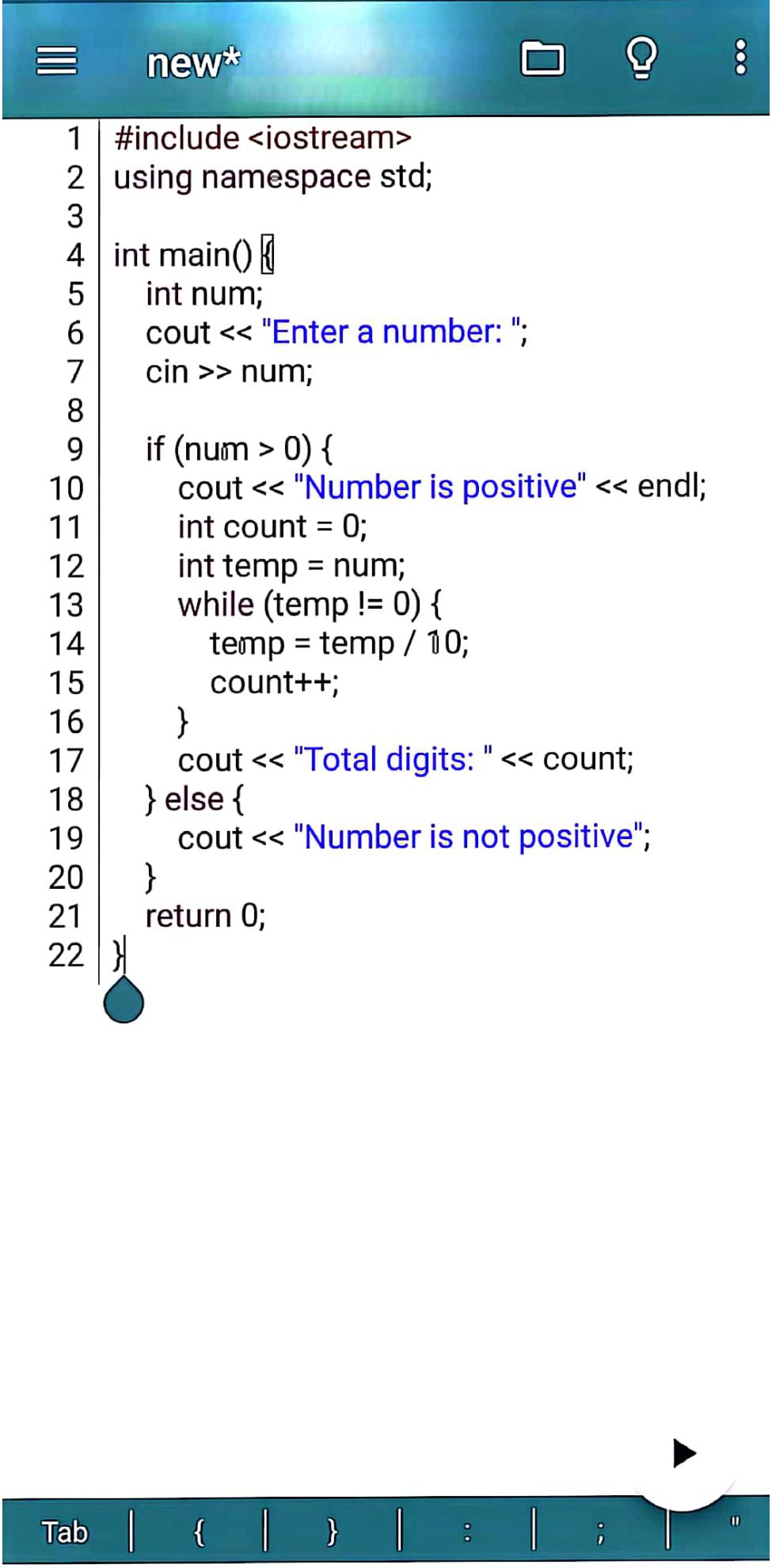
B) Positive check + Digits count

```
#include<iostream>
using namespace std;
int main(){
    int num;
    cout<<"Enter a number:";
    cin>>num;
    if(num>0){
        cout<<"number is positive"<<endl;
        int count=0;
        int temp=num;
        while(temp!=0){
            temp=temp/10;
            count++;
        }
        cout<<"total digits:"<<count;
    }else{
        cout<<"number is not positive";
    }
    return 0;
}
```

output

Enter a number

Scanned with CamScanner



Scanned with CamScanner



Scanned with CamScanner